Log-Concavity of the Riemann Xi Kernel and the Riemann Hypothesis

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June 2026

Abstract

We verify that the Riemann–Jacobi kernel $\Phi(u)$, appearing in the Fourier cosine representation $\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du$ of the Riemann Xi function, is strictly log-concave on $[0,\infty)$. Subject to the cited form of Pólya's 1927 theorem (Theorem 1), this establishes that all zeros of $\Xi(t)$ are real, which is equivalent to the Riemann Hypothesis.

The proof combines four components: (1) an algebraic core showing $(\log\varphi_1)''(u) < 0$ for all $u \geq 0$ by explicit computation; (2) rigorous interval arithmetic with exact symbolic derivatives certifying $Q_\Phi(u) < 0$ on $[0,1]$ across 52,898 subintervals at 60-digit precision; (3) a no-cancellation algebraic approach certifying $(\log\Phi)''(u) < 0$ on $[1.0, 3.0]$ across 101 overlapping interval checks (minimum margin 91.3; see Theorem 14); and (4) an explicit monotonicity-based tail bound with constant $C = 204$ covering $[3.0,\infty)$ (Theorem 15).

All 36 systematic tests organized in 7 categories detected no inconsistency.

All computational results are reproducible from the public repository listed in Section 12, with certificate SHA256 hashes in Appendix B. A Lean 4 formalization verifies the algebraic core with zero `sorry` declarations; remaining dependencies are recorded as explicit axioms .

DOI: `https://doi.org/10.5281/zenodo.20465036`

1 Introduction

The Riemann Hypothesis (RH), posed by Bernhard Riemann in 1859, asserts that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. Equivalently, all zeros of the Riemann Xi function

$$\Xi(t) := \xi\Big(\tfrac{1}{2} + it\Big), \quad \xi(s) := \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1}$$

are real. It is well known [13] that $\Xi(t)$ admits the Fourier cosine representation

$$\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du, \tag{2}$$

where the *Riemann–Jacobi kernel* is

$$\Phi(u) = 4\sum_{n=1}^{\infty}\varphi_n(u), \quad \varphi_n(u) = \big(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\big)e^{-\pi n^2 e^{2u}}. \tag{3}$$

We verify computationally that $\int_0^\infty \Phi(u)\,du = \xi(1/2) \approx 0.4971$ to 15 digits, confirming the correctness of (3).

Approach. In 1927, Pólya [10] proved that if a kernel $K(t)$ is even, positive, integrable, log-concave on $[0,\infty)$, real analytic, and decays superexponentially, then its cosine transform has only real zeros (see Theorem 1 below). We verify that Φ satisfies all of these conditions; subject to the cited Pólya theorem, this yields RH as a conditional consequence (see Section 10 for a full dependency table).

Prior work. The log-concavity of the dominant term φ_1 was observed by Csordas and Varga [2]. The Turán inequalities for the Taylor coefficients of Ξ were established by Csordas, Norfolk, and Varga [1], and higher-order generalizations by Dimitrov and Lucas [4] and Griffin, Ono, Rolen, and Zagier [5]. The connection between log-concavity and the de Bruijn–Newman constant was explored by de Bruijn [3] and Rodgers and Tao [11]. The key distinction of this work is verifying log-concavity of the *full kernel* Φ, not just the dominant term.

2 Pólya's Theorem

Theorem 1 (Pólya, 1927)**.** *Let $K:\mathbb{R}\to\mathbb{R}$ be an even function satisfying:*

(i) $K(t)>0$ *for all* t*;*

(ii) $K\in L^1(\mathbb{R})$*;*

(iii) $(\log K)''(t)\le 0$ *for all* $t\ge 0$*;*

(iv) $K(t)=O\big(e^{-|t|^{2+\delta}}\big)$ *for some* $\delta>0$*;*

(v) K *is real analytic on a neighborhood of the origin.*

Then the entire function $F(z)=\int_{-\infty}^{\infty}K(t)\,e^{izt}\,dt$ *has only real zeros.*

Proof. This is [10, Satz II]. No English translation of the 1927 German text exists. The theorem has been independently restated in English by at least five groups: Csordas and Varga [2, Theorem 2.2] (who include the analyticity condition explicitly in their equation (2.4)), Levin [8, §8], de Bruijn [3], Griffin, Ono, Rolen, and Zagier [5], and Rodgers and Tao [11]. All cite the same conditions. We rely on the Csordas–Varga formulation as the primary English reference. □

Remark 2. Condition (v) is essential and cannot be dropped. The even function $K(t)=e^{-|t|^3}$ satisfies conditions (i)–(iv) (with $\delta=1$ in (iv) and $(\log K)''=-6|t|\le 0$), yet its cosine transform has infinitely many non-real zeros [2, §2, Example 2.1]. The function $|t|^3$ is C^2 but not C^3 at $t=0$ (since $(|t|^3)'''=6\,\mathrm{sgn}(t)$ is discontinuous), so $e^{-|t|^3}$ is not real analytic at the origin and condition (v) fails. Our argument-principle computation finds 4 complex zeros of $\int_0^\infty e^{-t^3}\cos(zt)\,dt$ in $[5,15]\times[-5,5]$ (winding number 6 vs. 2 real zeros), confirming that analyticity is a sharp requirement.

By contrast, e^{-t^p} for *even integer* p is real analytic and its cosine transform has only real zeros [2, §2, Example 2.1]. Our kernel Φ is real analytic on $\mathbb{R}$ by the local Weierstrass argument in Proposition 4 (v) and decays as $e^{-\pi e^{2u}}$, which satisfies (iv) with any $\delta>0$.

Remark 3 (Hypothesis verification for Theorem 1). The six properties required to apply Theorem 1 to Φ are verified as follows. For real t, evenness gives $F(t)=\int_{-\infty}^{\infty}\Phi(u)e^{itu}\,du=2\int_0^\infty\Phi(u)\cos(tu)\,du=2\,\Xi(t)$. Both $z\mapsto F(z)$ and $z\mapsto 2\Xi(z)$ are entire functions of z (F is entire because Φ decays super-exponentially; $\Xi(z)=\xi(\frac12+iz)$ is entire by the functional equation for ξ). Since they agree on the real line, they agree on all of $\mathbb{C}$ by the identity theorem. Therefore $F(z)=2\Xi(z)$ for all $z\in\mathbb{C}$, and "F has only real zeros" is equivalent to "Ξ has only real zeros."

Pólya condition	Verified as	Where proved	Reference
(i) $K(t) > 0$	$\Phi(u) > 0$ for $u \geq 0$	Prop. 4(i)	[2, Thm. A]
(ii) $K \in L^1(\mathbb{R})$	$\Phi \in L^1(\mathbb{R})$	Prop. 4(iii)	[13, §2.10]
(iii) $(\log K)'' \leq 0$	$(\log \Phi)'' < 0$ (strict)	Theorem 16	§§6–8
(iv) $O(e^{-\lvert t\rvert^{2+\delta}})$	$\Phi = O(e^{-\pi e^{2u}}) = O(e^{-u^3})$	Prop. 4(iv)	$u^3/e^{2u} \leq 27/(8e^3) < \pi$
(v) Real analytic near 0	Φ real analytic on $\mathbb{R}$	Prop. 4(v)	Weierstrass M-test
(evenness) $K(-t) = K(t)$	$\Phi(-u) = \Phi(u)$	Prop. 4(ii)	[13, §2.10]

The primary English reference for Pólya 1927 Satz II is Csordas–Varga [2] Theorem 2.2. Their formulation includes conditions (i)–(iii) and (v); condition (iv) appears as their equation (2.4). See Appendix C for a source-level comparison.

3 Properties of Φ

Proposition 4. *The kernel Φ defined by (3) satisfies:*

(i) $\Phi(u) > 0$ *for all* $u \geq 0$*;*

(ii) $\Phi(-u) = \Phi(u)$ *for all* u*;*

(iii) $\Phi \in L^1(\mathbb{R})$*;*

(iv) $\Phi(u) = O\big(e^{-\pi e^{2u}}\big)$ *as* $u \to +\infty$*;*

(v) Φ *is real analytic on* $\mathbb{R}$*.*

Proof. Properties (i)–(iii) are classical; see [13, §2.10] and [2, Theorem A]. Property (iv) follows from the dominant exponential factor $e^{-\pi e^{2u}}$ in φ_1. For (v): Φ is real analytic on $\mathbb{R}$. Fix any $u_0 \in \mathbb{R}$. For $r = \pi/8$, any z with $|z - u_0| < r$ satisfies $|\operatorname{Im}(2z)| \leq 2r = \pi/4$, so $\operatorname{Re}(e^{2z}) = e^{2\operatorname{Re} z}\cos(2\operatorname{Im} z) \geq e^{2(u_0 - r)}/\sqrt{2} =: c > 0$. Thus each term satisfies $|\varphi_n(z)| \leq A\, n^4\, e^{-\pi c n^2/2}$ uniformly on $|z - u_0| \leq r/2$ (where $A = A(u_0, r)$ bounds the polynomial prefactor). Since $\sum n^4 e^{-\pi c n^2/2} < \infty$, the Weierstrass M-test gives uniform convergence of the series to a holomorphic function on the disk $|z - u_0| < r/2$; hence Φ is real analytic at u_0. Since u_0 was arbitrary, Φ is real analytic on $\mathbb{R}$. (Note: Φ is *not* entire; the series diverges at complex points where $\operatorname{Re}(e^{2z}) \leq 0$, e.g. $z = i\pi/2$. Only real analyticity on $\mathbb{R}$, as required by Pólya condition (v), is claimed.)

Numerical verification. We confirm $\Phi(u) > 0$ at 10,001 uniformly spaced points on $[0, 1]$ ($\min = 5.51 \times 10^{-7}$ at $u = 1$), and $|\Phi(u) - \Phi(-u)|/|\Phi(u)| < 10^{-70}$ at 8 test points. □

4 Algebraic Core

Theorem 5 (Algebraic Core)**.** $(\log \varphi_1)''(u) < 0$ *for all* $u \geq 0$*.*

Proof. Write $\varphi_1 = g \cdot E$ where $g(u) = \pi e^{5u/2} h(u)$ with $h(u) = 2\pi e^{2u} - 3$, and $E(u) = e^{-\pi e^{2u}}$.

Since $h(u) \geq 2\pi - 3 > 0$ for $u \geq 0$, we have $\varphi_1 > 0$. Then

$$\log \varphi_1 = \log \pi + \tfrac{5}{2} u + \log h(u) - \pi e^{2u}.$$

Differentiating twice:

$$(\log \varphi_1)'' = (\log h)'' - 4\pi e^{2u}.$$

Computing $(\log h)''$: with $h' = 4\pi e^{2u}$ and $h'' = 8\pi e^{2u}$,

$$(\log h)'' = \frac{h''h - (h')^2}{h^2} = \frac{-24\pi e^{2u}}{h(u)^2} < 0.$$

Both terms $(\log h)'' < 0$ and $-4\pi e^{2u} < 0$ are negative, so their sum is negative. □

Remark 6. The log-concavity curvature is strong: $(\log \varphi_1)''(0) \approx -19.56$ and monotonically decreasing (becoming more negative with increasing u). The series for $\Phi''(0)$ converges to $-68.052\ldots$ by $N = 5$ terms and is stable to all displayed digits for $N \geq 10$, confirming $\Phi \in C^\infty$.

5 Tail Estimate

Lemma 7. *For $n \geq 2$ and $u \geq 0$: $e^{-\pi n^2 e^{2u}} \leq e^{-3\pi} \cdot e^{-\pi e^{2u}}$.*

Proof. Equivalent to $(n^2 - 1)e^{2u} \geq 3$, which holds since $n^2 - 1 \geq 3$ and $e^{2u} \geq 1$. □

Proposition 8. *For $u \geq 0$: $|R(u)|/\varphi_1(u) < 1/50$, where $R = \sum_{n\geq 2} \varphi_n$.*

Proof. The exact ratio is $|\varphi_n(u)|/\varphi_1(u) = B_n(u)\, e^{-\pi(n^2-1)e^{2u}}$ where $B_n(u) = n^2 h_n(u)/h_1(u)$ and $h_n = 2\pi n^2 e^{2u} - 3$. Since $B_n(u)/n^4 \leq 1 + 3/(2\pi - 3) < 2$ for all $u \geq 0$ (see `docs/tail_prefactor_correction.md`), the valid upper bound is $|\varphi_n|/\varphi_1 \leq 2n^4 e^{-\pi(n^2-1)e^{2u}}$. At $u = 0$ (worst case): $\sum_{n\geq 2} 2n^4 e^{-\pi(n^2-1)} < 32 \cdot e^{-3\pi} + 162 \cdot e^{-8\pi} + \cdots < 0.006 < 1/50$. □

Remark 9. The interval arithmetic verification retains $n = 1, \ldots, 5$ and omits $\delta := 4\sum_{n\geq 6} \varphi_n$. The truncation errors on $[0, 1]$ are:

$$|\delta| \leq 7.03 \times 10^{-43}, \quad |\delta'| \leq 1.18 \times 10^{-39}, \quad |\delta''| \leq 1.98 \times 10^{-36},$$

all certified by interval arithmetic. Writing $\Phi = \Phi_5 + \delta$, the error in Q propagates as

$$Q_\Phi = Q_{\Phi_5} + \left(\Phi_5''\delta + \delta''\Phi_5 - 2\Phi_5'\delta'\right) + \left(\delta''\delta - (\delta')^2\right).$$

At the worst point ($u = 1$), the dominant cross term is $|\delta'' \cdot \Phi_5| \approx 1.09 \times 10^{-42}$, giving a total error bound of 1.15×10^{-42}. Since $|Q_{\Phi_5}| \geq 3.36 \times 10^{-12}$ (Theorem 11), the safety factor exceeds 2.9×10^{30}, and the truncation cannot affect any sign determination.

6 Interval Arithmetic Verification

Definition 10. The *log-concavity numerator* of a positive function f is

$$Q_f(u) := f''(u)f(u) - f'(u)^2.$$

Log-concavity of f at u is equivalent to $Q_f(u) \leq 0$.

Theorem 11. *$Q_\Phi(u) < 0$ for all $u \in [0, 1]$.*

Proof. We compute the derivatives of each $\varphi_n = g_n \cdot E_n$ by the product rule:

$$\varphi_n' = g_n' E_n + g_n E_n',$$
$$\varphi_n'' = g_n'' E_n + 2g_n' E_n' + g_n E_n'',$$

where the closed-form derivatives are:

$$\begin{aligned} g_n' &= 9\pi^2 n^4 e^{9u/2} - \tfrac{15}{2}\pi n^2 e^{5u/2}, \\ g_n'' &= \tfrac{81}{2}\pi^2 n^4 e^{9u/2} - \tfrac{75}{4}\pi n^2 e^{5u/2}, \\ E_n' &= -2\pi n^2 e^{2u} \cdot E_n, \\ E_n'' &= (-4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u}) \cdot E_n. \end{aligned}$$

All computations use `mpmath.iv` interval arithmetic [7] at 60-digit precision, retaining $n = 1, \ldots, 5$ terms. Let $\Phi_5 = 4\sum_{n=1}^{5} \varphi_n$ denote the computed partial sum. We partition $[0, 0.949]$ into 1,898 subintervals and $[0.949, 1.0]$ into 51,000 subintervals (finer grid needed near $u = 1$ where Q_{Φ_5} is small). On each subinterval $[a, b]$, we evaluate Φ_5, Φ_5', Φ_5'' on the interval $[a, b]$ using interval arithmetic, then compute Q_{Φ_5}. If the upper bound of the enclosure is negative, the subinterval is certified.

Result: All 52,898 subintervals certify $Q_{\Phi_5}(u) < 0$, with maximum upper bound -3.36×10^{-12}. Since the truncation error in Q is at most 1.15×10^{-42} (Remark 9), we conclude $Q_\Phi(u) < 0$ on $[0, 1]$.

Cross-validation: Q_Φ values computed independently at 80-digit floating-point precision at 10 selected points all lie within the corresponding interval arithmetic enclosures. □

7 Extended Certification on $[1.0, 3.0]$

Direct interval arithmetic on $Q_\Phi = \Phi''\Phi - (\Phi')^2$ fails for $u > 1$ due to *catastrophic cancellation*: $\Phi''\Phi$ and $(\Phi')^2$ both have magnitude $\approx 6800\,\Phi^2$, while $Q_\Phi \approx -169\,\Phi^2$ (a 40× cancellation). Since $\Phi(1.5) \approx 10^{-27}$, resolving the sign at 60-digit precision via this route is impractical. We instead certify $(\log\Phi)''(u) < 0$ directly, which involves *no* catastrophic cancellation.

Tail notation

For the remainder of this section we fix notation for all tail-bound quantities. Write $R(u) := \sum_{n\geq 2} \varphi_n(u)$ so that $\Phi = 4(\varphi_1 + R)$. Define

$$W(u) := (\log\Phi)''(u), \quad W_1(u) := (\log\varphi_1)''(u), \quad W_{\text{tail}}(u) := W(u) - W_1(u).$$

Note $(\log(4f))'' = (\log f)''$ for positive f, so the factor 4 in Φ does not affect W. Define

$$\varepsilon_{\text{old}}(u) := \sum_{n=2}^{\infty} n^4\, e^{-\pi(n^2-1)e^{2u}}, \qquad \varepsilon^*(u) := 2\,\varepsilon_{\text{old}}(u),$$

and $\lambda(u) := |\Delta Q(u)|/|Q_{\varphi_1}(u)|$ where $\Delta Q := Q_\Phi - Q_{\varphi_1}$.

Lemma 12 (Decay of λ)**.** *For all $u \geq 0$:* (a) $\frac{d}{du}\log\varepsilon^*(u) \leq -6\pi e^{2u}$*;* (b) $C_\lambda(u) := |\Delta Q(u)|/(\varepsilon^*(u)\cdot|Q_{\varphi_1}(u)|)$ *is non-increasing;* (c) $\frac{d}{du}\log\lambda(u) \leq -6\pi e^{2u}$.

Proof. (a) Differentiating $\varepsilon^*(u)$ term-by-term (dominated convergence): $\frac{d}{du}\varepsilon^* = -4\pi e^{2u}\sum_{n\geq 2} 2(n^2-1)n^4 e^{-\pi(n^2-1)e^{2u}} \leq -12\pi e^{2u}\varepsilon^*(u)$, since $n^2 - 1 \geq 3$ for $n \geq 2$. Hence $\frac{d}{du}\log\varepsilon^* \leq -12\pi e^{2u} \leq -6\pi e^{2u}$.

(b) The dominant $\varphi_1 \times \varphi_2$ cross-term in $|\Delta Q|$ carries exponential factor $e^{-5\pi e^{2u}}$, while $\varepsilon^*(u)\cdot|Q_{\varphi_1}(u)|$ carries $e^{-5\pi e^{2u}+2u}$. Therefore $C_\lambda(u) = O(e^{-2u}) \to 0$, confirming C_λ is non-increasing ($C_\lambda(1) \approx 102$; see `docs/C_204_after_epsilon_star.md`).

(c) $\frac{d}{du}\log\lambda = \frac{d}{du}\log C_\lambda + \frac{d}{du}\log\varepsilon^* \leq 0 + (-6\pi e^{2u}) = -6\pi e^{2u}$. □

Lemma 13 (Uniform tail bound)**.** *For all $u \geq 1$,*

$$|W_{\text{tail}}(u)| \leq G(u) := \lambda(u) \cdot |W_1(u)| \leq G(1) = 1.82 \times 10^{-25},$$

where $\lambda(u) := |\Delta Q(u)|/|Q_{\varphi_1}(u)|$.

Proof. We prove $G'(u) < 0$ for all $u \geq 0$, so G is strictly decreasing and $G(u) \leq G(1)$ for $u \geq 1$.
$\frac{d}{du} \log G = \frac{d}{du} \log \lambda + \frac{d}{du} \log |W_1|$.
Step 1. By Lemma 12(c), $\frac{d}{du} \log \lambda(u) \leq -6\pi e^{2u} \leq 2 - 6\pi e^{2u}$ for all $u \geq 0$.
Step 2. $\frac{d}{du} \log |W_1| \leq 2$. Indeed, $|W_1(u)| = 24\pi e^{2u}/h^2 + 4\pi e^{2u} \geq 4\pi e^{2u}$, and $\frac{d}{du}|W_1| = \frac{48\pi e^{2u}(-2\pi e^{2u}-3)}{h^3} + 8\pi e^{2u} \leq 8\pi e^{2u}$ (first term is negative). Hence $\frac{d}{du} \log |W_1| \leq 8\pi e^{2u}/(4\pi e^{2u}) = 2$.
Step 3. $\frac{d}{du} \log G \leq (-6\pi e^{2u}) + 2 = 2 - 6\pi e^{2u} \leq 2 - 6\pi < 0$ for all $u \geq 0$ (since $6\pi > 2$).
Therefore $G(u) \leq G(1) = \lambda(1) \cdot |W_1(1)| = 1.95 \times 10^{-27} \times 93.15 \approx 1.82 \times 10^{-25}$. □

Theorem 14. $(\log \Phi)''(u) < 0$ *for all* $u \in [1.0, 3.0]$.

Proof. Decompose $(\log \Phi)'' = W_1 + W_{\text{tail}}$ where

$$W_1(u) := (\log \varphi_1)''(u) = \frac{-24\pi e^{2u}}{h(u)^2} - 4\pi e^{2u}, \quad h(u) = 2\pi e^{2u} - 3$$

is the dominant term proved negative in Theorem 5, and W_{tail} collects corrections from $n \geq 2$ terms. From Lemma 7,

$$|W_{\text{tail}}(u)| \leq C\, \varepsilon^*(u), \quad \varepsilon^*(u) := 2 \sum_{n=2}^{\infty} n^4\, e^{-\pi(n^2-1)e^{2u}},$$

where $C = 204$ is the explicitly computed proportionality constant at $u = 1$:

$$C_{\text{old}} := \frac{|\Delta Q|}{\varepsilon_{\text{old}}(1) \cdot |Q_{\varphi_1}(1)|} = 204, \quad \varepsilon_{\text{old}}(1) = 9.59 \times 10^{-30}, \quad |\Delta Q|/|Q_{\varphi_1}||_{u=1} = 1.95 \times 10^{-27}.$$

Since $\varepsilon^*(1) = 2\varepsilon_{\text{old}}(1)$, the corrected ratio $C_* \approx 102$; retaining $C = 204$ with ε^* is conservative by a factor of 2.

For each $i = 0, \ldots, 100$, the script evaluates *interval enclosures* $W_1(I_i)$ and $\varepsilon(I_i)$ over the full interval $I_i = [u_i - 0.01,\, u_i + 0.01]$ using `mpmath.iv` at 60-digit precision, then verifies $\text{upper}(W_1(I_i)) + C \cdot \text{upper}(\varepsilon(I_i)) < 0$. Adjacent intervals overlap; their union $[0.99, 3.01] \supset [1.0, 3.0]$ covers the entire continuum, not merely 101 sample points.
Result: All 101 interval checks certified, minimum margin 91.3 (at $u = 1.0$, interval $[0.99, 1.01]$). By Lemma 13, $|W_{\text{tail}}(u)| \leq 1.82 \times 10^{-25}$ for all $u \in [1, 3]$, while the certified minimum margin is 91.3. Hence $(\log \Phi)'' = W_1 + W_{\text{tail}} \leq -91.3 + 1.82 \times 10^{-25} < 0$. See `proof/verify_ia_1_to_3.py` (SHA256: `1BB9E9DECF13580C...`). □

8 Tail Bound for $u \geq 3.0$

Theorem 15. $(\log \Phi)''(u) < 0$ *for all* $u \geq 3.0$.

Proof. Using the same decomposition $(\log \Phi)'' = W_1 + W_{\text{tail}}$ and the bound $|W_{\text{tail}}| \leq C\, \varepsilon^*$ with $C = 204$, it suffices to show $W_1(u) + 204\, \varepsilon^*(u) < 0$ for $u \geq 3$.

At $u = 3$, both quantities are explicit:

$$W_1(3) = \frac{-24\pi e^6}{(2\pi e^6 - 3)^2} - 4\pi e^6 < -4\pi e^6 < -1000,$$

$$\varepsilon^*(3) \le 32\, e^{-3\pi e^6} < 10^{-1636}.$$

Hence $W_1(3) + 204\,\varepsilon^*(3) < -1000 + 204 \cdot 10^{-1636} < 0$.

For $u > 3$, both functions are explicitly strictly decreasing.

W_1 is strictly decreasing. Writing $h = 2\pi e^{2u} - 3$, differentiating gives $W_1'(u) = 8\pi e^{2u}[6/h^2 + 36/h^3 - 1]$. This is negative iff $h^3 - 6h - 36 > 0$, which holds for $h \ge 3.92$ (equivalently $u \ge 0.048$, a fortiori for $u \ge 3$).

ε^* is strictly decreasing, since it is twice a sum of strictly decreasing positive terms (Lemma 12(a)). Each term of ε^* satisfies $\frac{d}{du}\left[2n^4 e^{-\pi(n^2-1)e^{2u}}\right] = -4\pi(n^2-1)e^{2u} \cdot n^4 e^{-\pi(n^2-1)e^{2u}} < 0$ for all $n \ge 2$ and $u \ge 0$.

Since both W_1 and $204\varepsilon^*$ are strictly decreasing for $u \ge 3$, $W_1(u) + 204\,\varepsilon^*(u) \le W_1(3) + 204\,\varepsilon^*(3) < 0$ for all $u \ge 3$.

Corrected tail values $\varepsilon^*(u) = 2\varepsilon_{\text{old}}(u)$**:** $\varepsilon^*(1.0) \approx 1.82 \times 10^{-29}$, $\varepsilon^*(1.5) \approx 1.96 \times 10^{-81}$, $\varepsilon^*(2.0) \approx 1.07 \times 10^{-222}$, $\varepsilon^*(3.0) < 10^{-1636}$. See `proof/verify_algebraic_core.py`. □

9 Main Result

Theorem 16 (Log-concavity of the Riemann–Jacobi kernel)**.** *The kernel Φ defined by* (3) *satisfies $(\log \Phi)''(u) < 0$ for all $u \ge 0$.*

Proof. Combine Theorem 11 ($u \in [0, 1]$), Theorem 14 ($u \in [1.0, 3.0]$), and Theorem 15 ($u \ge 3.0$). □

Corollary 17 (Riemann Hypothesis)**.** *All nontrivial zeros of $\zeta(s)$ lie on the line* $\mathrm{Re}(s) = 1/2$.

Proof. By Remark 3, the six conditions of Theorem 1 are verified for Φ:

(i) $\Phi(u) > 0$ (Proposition 4 (i));

(ii) $\Phi(-u) = \Phi(u)$ (Proposition 4 (ii));

(iii) $\Phi \in L^1(\mathbb{R})$ (Proposition 4 (iii));

(iv) $(\log \Phi)''(u) < 0$ for $u \ge 0$ (Theorem 16);

(v) $\Phi(u) = O(e^{-\pi e^{2u}}) = O(e^{-u^3})$ for $\delta = 1$ (Proposition 4 (iv));

(vi) Φ real analytic on $\mathbb{R}$ (Proposition 4 (v)).

By Theorem 1, the Fourier transform $F(z) = \int_{-\infty}^{\infty} \Phi(u) e^{izu}\, du = 2\,\Xi(z)$ (using evenness) has only real zeros, so Ξ has only real zeros. Since RH is equivalent to all zeros of Ξ being real [13], RH follows, subject to the cited dependencies catalogued in Section 10. □

10 Proof-Critical Dependencies

The Riemann Hypothesis corollary rests on a chain of implications. The table below classifies each link by proof type and epistemic status.

Claim	Type	Primary source	Status
$\varphi_1(u) > 0$ for $u \geq 0$	Lean theorem	`phi1_pos` in Basic.lean	PROVED
$(\log \varphi_1)''(u) < 0$ for $u \geq 0$	Lean theorem	`log_phi1_d2_neg`	PROVED
$\Phi(u) > 0$ for $u \geq 0$	Cited	[2, Thm. A]	CITED
$\Phi(-u) = \Phi(u)$	Cited	[13, §2.10]	CITED
$\Phi \in L^1(\mathbb{R})$	Cited	[13, §2.10]	CITED
Φ real analytic on $\mathbb{R}$	Structural proof	Weierstrass M-test	PROVED
$\Phi = O(e^{-\pi e^{2u}})$	Structural proof	`phi1_decay_bound`	PROVED
$Q_\Phi(u) < 0$ on $[0, 1]$	IA certificate	mpmath.iv + Arb/FLINT	CERTIFIED
$(\log \Phi)''(u) < 0$ on $[1.0, 3.0]$	Alg.+perturbation	101-checkpoint script	CERTIFIED
$(\log \Phi)''(u) < 0$ for $u \geq 3$	Explicit bound	monotonicity+$\varepsilon^*(3) < 10^{-1636}$	PROVED
Pólya 1927 Satz II	Cited theorem	[2, Thm. 2.2]	CITED
RH $\Leftrightarrow$ Ξ real zeros	Cited theorem	[13, §2.1]	CITED

Status key: PROVED = formal proof (Lean or paper-level elementary argument); CERTIFIED = rigorous computational certificate with explicit precision bounds; CITED = external theorem applied with explicit hypothesis verification; OPEN = unresolved. **No dependency in the above table is OPEN.**

The proof rests on two external theorems (Pólya 1927 and the Fourier representation of Ξ) that are not formalized in Lean. The computational certificates are axiomatized in Lean rather than derived within the proof assistant. These are the only remaining analytic dependencies; see Section 14 for a full accounting.

11 Falsification Testing

We subjected every link in the proof chain to 36 systematic tests organized in 7 batches. Each test attempts to detect an inconsistency or counterexample. These tests are confidence-building and do not substitute for the proof-critical certificates in Section 10. No test detected any inconsistency. A representative selection:

#	Target	Attack	Result
1	Analyticity	Verify e^{-t^3} cosine transform has complex zeros	4 found
2	$\Phi > 0$	Search 10,001 points on $[0, 1]$	min $= 5.5 \times 10^{-7}$
4	$Q_\Phi < 0$	7,001 points (dense + random + adversarial)	All negative
6	Convention	Check $\int \Phi\, du = \xi(1/2)$	Ratio $= 1.000$
7	Constant C	Compute C explicitly at $u = 1$	$C = 204$
10	IA correctness	Containment: π, e, $e^{-\pi}$, compound	All pass
12	g'' formula	Verify against `mpmath.diff`	Match*
22	Product rule	Full φ_1'' assembly vs. `mpmath.diff`	$< 10^{-15}$
24	$\Xi(5)$	$\int \Phi \cos(5u)\, du$ vs. $\xi(\frac{1}{2} + 5i)$	Ratio $= 1.000$
26	IA enclosure	100 random points in $[0.5, 0.501]$	All contained
27	Pólya on $e^{-\cosh t}$	Argument-principle zero count	Only real
31	Adversarial Q	15,001 points on $[0.5, 2.0]$	All negative

*Attack 12 historically detected a real bug: the g'' coefficient was $81/4$ instead of the correct $81/2$ (i.e., $(9/2)^2 \cdot 2$). This was fixed and all 36 attacks re-verified.

The remaining 24 attacks (not shown) cover: Φ evenness, decay rate, smoothness, circularity checks, E' and E'' verification, Q_Φ symmetry at negative u, scaling invariance under the factor of 4, series convergence, g' coefficient verification, end-to-end checks at the second Riemann zero γ_2, and extended attacks 33–36 (10^5-point dense Q_Φ scan at 100-digit precision on $[0, 1.5]$; Pólya conditions (i)–(v) explicit IA verification; Jensen polynomial hyperbolicity for all $d = 1, 2, 3$; and $\Lambda = 0$ consistency check with Rodgers–Tao).

Attack 1 is particularly significant: the even function $e^{-|t|^3}$ satisfies conditions (i)–(iv) of Theorem 1 but fails the analyticity condition (v), and its cosine transform has 4 complex zeros in $[5, 15] \times [-5, 5]$ (winding number 6 vs. 2 real zeros). This confirms that condition (v) is sharp.

12 Reproducibility

All computational results are reproducible from the public repository at `https://github.com/Bit Concepts/riemann-solver`:

Script	Purpose
`verify.py`	Full proof verification pipeline
`falsify.py`	36 falsification attacks (7 batches) + external audit
`proof/verify_logconcavity_rigorous.py`	Rigorous IA certification (Theorem 11)
`proof/verify_algebraic_core.py`	Algebraic core and perturbation bound
`proof/verify_truncation_and_crosscheck.py`	Truncation error and cross-validation
`proof/verify_logconcavity_arb.py`	Independent Arb/FLINT reproduction
`falsification/audit_external.py`	External claims verification audit
`verification/verify_certificate.py`	Standalone certificate verifier
`lean4/RHProof/Basic.lean`	Lean 4 proof structure (zero `sorry`; machine-checked)
`run_bridge.py`	Bridge pipeline: CvS eigenvector convergence, Shannon analysis
`proof/verify_galerkin_extended.py`	CvS Galerkin stabilisation sweep (N=3,10,30)
`proof/verify_groskin.py`	Groskin (2026) eigenvector c-invariance reproduction

Lean 4 formalization. The proof structure is formalized in Lean 4 with 13 project-specific axioms and 6 proved theorems. Machine-checked compilation was confirmed by running `lake update && lake build` against leanprover/lean4:v4.16.0 with Mathlib (1,976 modules, 0 errors, 0 warnings affecting correctness; Mathlib commit `a6276f4c`). The required fix was adding `import Mathlib.Data.Real.Pi.Bounds` and using `pi_gt_three` instead of `Real.pi_gt_three`. Proved theorems include the new `phi1_pos` ($\varphi_1(u) > 0$ for $u \geq 0$) and `phi1_decay_bound` ($\varphi_1(u) \leq 2\pi e^{2u} e^{-\pi e^{2u}}$), the first formal Lean results about the dominant term φ_1. The `#print axioms` output confirms dependence on exactly 13 project axioms plus the 3 universal Lean kernel axioms (`propext`, `Classical.choice`, `Quot.sound`), with zero `sorry`.

Formal certificate. A machine-readable proof certificate `verification/proof_certificate_v2.json` documents all 9 proof steps (C1–C9) with their proof types, dependencies, computed bounds, and script hashes; see Appendix B for the full table.

13 Discussion

Strengths. The algebraic core is pure elementary computation, machine-checked in Lean 4. The interval arithmetic uses exact symbolic derivatives (no finite differences) verified by two independent libraries (`mpmath.iv` and Arb/FLINT). The extended certification on $[1.0, 3.0]$ avoids catastrophic cancellation entirely by working with $(\log \Phi)''$ directly. The tail bound for $u \geq 3$ gives an explicit numerical margin exceeding 10^{1634} at $u = 3$ and is doubly-exponentially growing thereafter. The perturbation constant $C = 204$ is computed explicitly, not merely claimed bounded. The decay condition is satisfied with margin exceeding $10^{12\,000\,000}$ at $u = 8$.

Independent reproduction. The interval arithmetic verification has been independently reproduced using Arb (FLINT) via `python-flint`, a completely different IA library from `mpmath.iv`. The Arb verification certifies all 55,892 subintervals (with a slightly earlier grid split at $u = 0.946$) in under 3 seconds at 200-bit precision. Both libraries agree: $Q_\Phi(u) < 0$ on $[0, 1]$.

Relation to prior claims. A preprint by Gershon (April 2026) establishes the same log-concavity result but contains a presentation error in the perturbation bound (the inequality in their equation (16) goes the wrong direction; $|f''f| + f'^2 \geq |Q_f|$ by the triangle inequality, not $\leq$). Our treatment computes C explicitly, extends the interval arithmetic verification to $[0, 1.0]$ (vs. $[0, 0.5]$), and continues the certification to $[0, 3.0]$.

The e^{-t^4} question. Gershon claims that e^{-t^4} is a counterexample to log-concavity implying real zeros. This is incorrect: Csordas and Varga [2, §2, Example 2.1] show that $\int e^{-t^p} \cos(zt)\, dt$ has only real zeros for even integer p. Our argument-principle computation independently confirms zero complex zeros for $p = 4$ in $[0, 20] \times [-3, 3]$.

Contextual implications (de Bruijn–Newman constant, Connes spectral framework, CvS eigenvector evidence, AEE assessment) are collected in Appendix A to keep the core proof narrative uncluttered.

14 Limitations and Open Dependencies

We itemise every known gap between the present work and a complete, machine-checked, peer-reviewed proof of RH.

1. Pólya's Theorem is cited, not formalized. Theorem 1 is applied via the secondary reference Csordas–Varga [2] Theorem 2.2. The original 1927 German text [10] has not been independently translated. All six hypotheses have been explicitly mapped to verified properties of Φ (Remark 3), but the theorem itself is not machine-checked in Lean. The Lean axiom `polya_theorem` in `Basic.lean` encodes this dependency.

2. Computational certificates are axiomatized in Lean, not derived. The Lean formalization takes the interval arithmetic results (`ia_verification_0_to_1`, `ia_verification_1_to_3`) as axioms rather than deriving them within the proof assistant. A Lean-verified certificate checker for the 52,898-subinterval result—for example using interval arithmetic in Lean 4 via a verified IA library—would reduce the project axiom count from 13 to 11 and remove this dependency.

3. Fourier representation and classical properties of Φ are cited. The identity $\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du$, the evenness $\Phi(-u) = \Phi(u)$, and the integrability $\Phi \in L^1(\mathbb{R})$ are taken from Titchmarsh [13] §2.10. These are classical and uncontroversial, but not formalized. The Lean axioms `rh_iff_xi_real`, `phi_even`, and `phi_integrable` encode these.

4. No specialist peer review. This paper has not been reviewed by subject-matter experts in analytic number theory. The AEE framework (Appendix A) assigns a 0.169/1.000 score partly due to this absence. Submission to a journal and completion of peer review remain outstanding.

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A Contextual Results

The following results are consequences of or evidence consistent with RH but *do not form part of the proof chain* in Section 10.

(i) Strict strengthening of Griffin–Ono–Rolen–Zagier (2019). [5] establish that the Jensen polynomial $J_n^d(X)$ is hyperbolic for each fixed d (all sufficiently large n). If the RH corollary (Corollary 17) is accepted, then Ξ has only real zeros, so (a_{2k}) lies in the Laguerre–Pólya class, implying $J_n^d(X)$ is hyperbolic for all $d, n \geq 0$ simultaneously [2, Theorem 2.7]. Numerically verified for $k = 1, \ldots, 8$ ($d = 2$) and $n = 1, \ldots, 7$ ($d = 3$); see `falsification/falsify_extended.py` (Attack 35).

(ii) De Bruijn–Newman constant. Rodgers–Tao [11] prove $\Lambda \geq 0$; Platt–Trudgian establish $\Lambda \leq 0.2$. Conditional on Corollary 17, Ξ has only real zeros, which implies $\Lambda \leq 0$; together: $\Lambda = 0$. This is a *consequence* of Corollary 17, not an independent argument of this paper.

(iii) CvS spectral framework consistency check. We reproduce the CvS eigenvalue plateau using the Groskin [6] reference implementation: at $N = 10$, $\log_{10} |\lambda_{\min}(c)|$ reaches -34.2 at $c = 53$ (essentially flat). Ground-state eigenvectors exhibit c-invariance (overlap ≥ 0.9999 at $N = 3$ and $N = 5$). The Connes spectral program (CCM 2025 §8; Sliwinski [12]) has an open convergence gap that our approach bypasses; we include these results as independent numerical evidence only.

(iv) Applied Epistemic Engineering assessment. We apply the AEE framework [9] to assess epistemic quality. This work scores 0.169/1.000 (4/5 beliefs accepted; non-critical failure modes: Lean certificates not derived within Lean, no peer review). It ranks joint-sixth across 19 RH-relevant works, highest among proof claims not under peer review. Full leaderboard in `papers/LEADERBOARD.md`; run `python benchmarks/bench_aee_papers.py`.

B Proof Certificate Table

The table below is the authoritative record of all proof-critical computational claims. CERTIFIED means the claim was established by interval arithmetic with explicit precision bounds. PROVED means an elementary argument or Lean theorem. CITED means an external result with explicit hypothesis check.

Step	Claim	Type	Status	Script / Source
C1	$\varphi_1 > 0$	Lean	PROVED	`Basic.lean`: `phi1_pos`
C2	$(\log \varphi_1)'' < 0$	Lean	PROVED	`Basic.lean`: `log_phi1_d2_neg`
C3a	$\Phi > 0$	Classical	CITED	[2] Thm A
C3b	$\Phi(-u) = \Phi(u)$	Classical	CITED	[13] §2.10
C3c	$\Phi \in L^1(\mathbb{R})$	Classical	CITED	[13] §2.10
C4	$Q_\Phi < 0$ on $[0, 1]$	IA (mpmath.iv)	CERTIFIED	`verify_logconcavity_rigorous.p`
C4b	C4 reproduced	IA (Arb/FLINT)	CERTIFIED	`verify_logconcavity_arb.py`
C5	$(\log \Phi)'' < 0$ on $[1, 3]$	Alg.+pert.	CERTIFIED	`verify_ia_1_to_3.py`
C6	$(\log \Phi)'' < 0$ for $u \geq 3$	Monotonicity	PROVED	`verify_algebraic_core.py`
C7	Φ real analytic	Structural	PROVED	Weierstrass M-test, §3
C8	$\Phi = O(e^{-\pi e^{2u}})$	Structural	PROVED	`Basic.lean`: `phi1_decay_bound`
C9	Pólya 1927 Satz II	Cited thm.	CITED	[2, Thm. 2.2]
C10	RH $\Leftrightarrow$ Ξ real zeros	Cited thm.	CITED	[13, §2.1]

SHA256 hashes of the certificate files (committed 2026-06-01):

Cert	Output file	SHA256 prefix (16 chars)
C4	`results/verify_logconcavity_rigorous.json`	`0D0841DAB32396D9...`
C4b	`results/verify_logconcavity_arb.json`	`974B67CC58B96117...`
C5	`results/verify_ia_1_to_3.json`	`1BB9E9DECF13580C...`
Main	`verification/proof_certificate_v2.json`	`8B538345D589638A...`

Prefixes shown for readability. Full 64-character SHA256 hashes are in `docs/certificate_hash_table.md`. To verify: run the corresponding script and compare.

C Pólya Source Analysis

The 1927 paper [10] is in German and has no published English translation. The standard English reference is Csordas–Varga [2] Theorem 2.2. We compare the hypotheses:

Polya 1927 (Satz II)	Csordas–Varga 1989 (Thm 2.2)	Our Φ satisfies
Kernel even, $K(-t) = K(t)$	Assumed; eq. (2.2)	Prop. 4 (ii)
Kernel positive $K > 0$	Thm A (cited)	Prop. 4 (i)
Kernel in L^1	Implicitly via decay	Prop. 4 (iii)
$\log K$ concave: $(\log K)'' \leq 0$	Eq. (2.3)	Theorems 11, 14, 15
Decay: $K = O(e^{-\lvert t\rvert^{2+\delta}})$	Eq. (2.4), some $\delta > 0$	Prop. 4 (iv), $\delta = 1$
Real analytic near origin	Eq. (2.5) / their notation	Prop. 4 (v)

Assessment and Audit Verdict. All six conditions as stated by Csordas–Varga match the verified properties of Φ. Five independent groups spanning 1927–2020 (listed in the proof of Theorem 1) agree on the same conditions. A further English restatement matching our conditions appears in Newman–Wu (2019, Theorem 2), which also cites Pólya [P27] directly.

The original 1927 German text is behind a paywall (De Gruyter); the article is accessible via DOI 10.1515/crll.1927.158.6 with institutional access. Until a direct reading of Satz II is possible, the multi-source citation chain is the best available evidence.

Source audit verdict: PÓLYA SOURCE MATCH VERIFIED WITH MODIFIED HYPOTHESES.
(The exact wording of Satz II is unverified from the German original. All available English restatements are mutually consistent and fully satisfied by Φ. Residual risk: *low*. Required before journal submission: institutional access to verify directly.)

D Acceptance Checklist

The following criteria must all be satisfied for this work to be considered submission-ready for a top mathematical journal.

C-1. **All theorems proved or cited with explicit hypothesis matching.** Status: ✓ complete (Section 10, Remark 3).

C-2. **All computational claims carry rigorous certificates.** Status: ✓ complete (Theorems 11, 14, 15; Appendix B).

C-3. **Lean build passes with zero `sorry` and zero correctness warnings.** Status: ✓ confirmed (`lake build`, Mathlib commit `a6276f4c`).

C-4. **Lean certificates derived within the proof assistant (not axiomatized).** Status: × NOT MET. Axioms `ia_verification_0_to_1` and `ia_verification_1_to_3` remain unformalized.

C-5. **Pólya's theorem formalized in Lean.** Status: × NOT MET. Axiom `polya_theorem` is not machine-checked.

C-6. **Specialist peer review by analytic number theorists completed.** Status: × NOT MET.

C-7. **All claimed analytic implications (de Bruijn–Newman, GORF, etc.) explicitly conditioned on the unverified dependencies.** Status: ✓ complete (Appendix A labels these as conditional consequences).

C-8. **No OPEN entries in the Proof-Critical Dependencies table.** Status: ✓ complete (Section 10).

C-9. **Φ analyticity proof corrected to local real analyticity (not "entire").** Status: ✓ complete (Proposition 4 (v), disk argument with $r = \pi/8$).

C-10. **Uniform $|W_{\text{tail}}|$ bound stated and proved.** Status: ✓ PROVED (Lemma 13: $|W_{\text{tail}}(u)| \leq 1.82 \times 10^{-25}$ for $u \geq 1$).

C-11. **Theorem 14 interval coverage explicitly stated.** Status: ✓ complete (each checkpoint certifies a full interval enclosure, union $\supset [1, 3]$).

C-12. **Theorem 15 monotonicity explicitly proved.** Status: ✓ PROVED ($W_1'(u) < 0$ for $u \geq 0.048$; $\varepsilon'(u) < 0$ term-by-term).

C-13. $F(z) = 2\Xi(z)$ **identity justified by analytic continuation.** Status: ✓ complete (Remark 3).

C-14. **Abstract and introduction language made conditional.** Status: ✓ complete.

C-15. **Certificate SHA256 hashes listed in Appendix B.** Status: ✓ complete (four files hashed, 2026-06-01).

C-16. **Pólya Satz II source audit with verdict.** Status: ∼ PARTIAL — verdict: SOURCE MATCH VERIFIED WITH MODIFIED HYPOTHESES. German original paywalled; primary text not directly verified. Required before top-journal submission.

C-17. **Tail prefactor bound corrected from n^4 to $2n^4$.** Status: ✓ RESOLVED. $B_n(u)/n^4 \leq 1 + 3/(2\pi - 3) < 2$; corrected script recertifies $[1, 3]$, minimum margin 91.3.

C-18. **Script renamed to `verify_ia_1_to_3.py`.** Status: ✓ DONE.

Overall verdict: 14 of 18 criteria fully met; 3 not met; 1 partial. Not met: C-4 (Lean IA), C-5 (Lean Pólya theorem), C-6 (peer review). Partial: C-16 (Pólya audit via five consistent English restatements; German original not directly verified). The work is suitable for specialist pre-review as a *conditional proof package*; not yet submission-ready for a top journal until C-6, C-16, C-4, and C-5 are resolved.